

SECTION AAnswer **all** questions.

1. State why chlorine is used in water treatment. [1]

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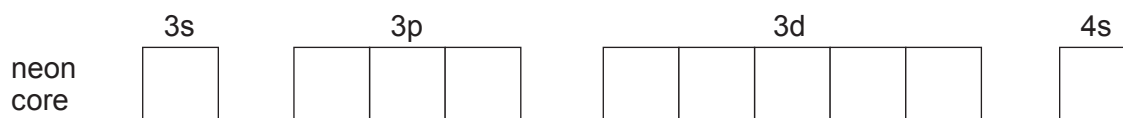
2. State the pattern in electronegativity across Period 3 in the Periodic Table. [1]

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3. Name the element in Period 3 that has a half-filled set of *p*-orbitals. [1]

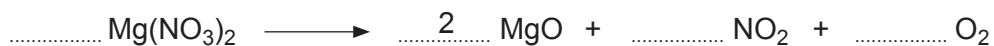
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4. By inserting arrows to represent electrons, complete the electronic structure of a vanadium atom. [1]



5. Magnesium nitrate decomposes when heated to form magnesium oxide, nitrogen dioxide and oxygen.

Balance the equation for this reaction. [1]



6. Consider the following species.



State whether you agree with the following statements, giving a reason for your conclusion.

(a) ${}^{32}_{16}\text{S}^{2-}$ has twice the number of neutrons that there are in ${}^{18}_8\text{O}$. [1]

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(b) The sum of the number of electrons in ${}^{16}_8\text{O}^{2-}$ and ${}^{18}_8\text{O}$ is equal to the number of electrons in ${}^{32}_{16}\text{S}^{2-}$. [1]

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7. Hydrated copper(II) sulfate has the formula $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.

Calculate the number of moles of oxygen atoms present in 0.5 mol of hydrated copper(II) sulfate. [1]

Oxygen atoms = mol

8. A 0.717 g sample of gas occupies 1000 cm^3 at 273 K and 1 atm. Calculate the relative formula mass of the gas. [2]

Relative formula mass =

